

Quantum Foundations of the Rope Model

A Non-Local Hidden-Variable Framework, and Its Confrontation with Bell's Theorem

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Naming note (5 September 2026). This programme is now the Mesh Programme and the theory it develops the mesh framework; 'the rope framework' in earlier revisions reads 'the mesh framework'. 'Rope' remains the name of the two-strand wound object, and the Rope Hypothesis credited to Bill Gaede keeps its name. Identifiers (repository, rope_solver, file names, claim IDs) are unchanged.

Abstract

Earlier papers show quantum mechanics EMERGING from the rope medium — the Dirac equation from two-strand mechanics, phase coherence, and decoherence. None confronts the interpretive core. If the two-strand imbalance is physically real and definite, as the programme's ontology insists, then the rope model is a HIDDEN-VARIABLE theory of quantum mechanics, and it must face Bell's theorem. This paper does so head-on. We show the confrontation forces a single honest conclusion: the rope model cannot be a LOCAL hidden-variable theory — that version is already falsified by loophole-free Bell experiments — and, because it can abandon neither realism (its reason to exist) nor measurement independence (without corroding its own falsifiability), it must be an explicitly NON-LOCAL hidden-variable theory, in the family of Bohmian mechanics. We argue the rope substrate supplies a natural non-local connection through shared/linked rope topology established at pair creation, and we are equally explicit about the two calculations that would confirm or kill the framework: deriving the exact quantum correlation $E(a,b) = -\cos(a-b)$ with the Tsirelson bound, and proving the connection respects no-signalling. We identify the model's central vulnerability — that a shared-rope correlation secretly fixed at the source would be a LOCAL hidden variable and hence already refuted — and treat avoiding it as a requirement, not a detail. The value of the paper is a definite, distinctive, falsifiable commitment where speculative theories usually equivocate.

What this paper claims — and does not

CLAIMS: that the rope model is necessarily a NON-LOCAL hidden-variable theory; that local rope hidden-variable models are already falsified; that the rope substrate offers a plausible non-local mechanism; and that two specific calculations decide the framework's fate.

DOES NOT CLAIM: to have DERIVED the quantum correlations from rope dynamics, to have PROVEN no-signalling, or to have solved the measurement problem. These are stated as open, and the paper is written to expose the model's vulnerabilities, not to hide them.

1. The Interpretive Core the Programme Has Avoided

The mesh programme has shown, in earlier papers, quantum-mechanical structure emerging from the medium: the Dirac equation from two-strand rope mechanics, phase coherence and its loss (decoherence), and quantum fluctuations about rope knots. These are results about the FORMALISM. They leave untouched the interpretive question that any realist substrate theory must eventually answer: what is a quantum measurement, and how can a theory with definite underlying values reproduce the correlations that Bell's theorem shows no local such theory can?

This is not optional decoration. It is the point at which the rope model is most testable and most vulnerable. A theory that posits a real, definite, physical imbalance underlying quantum states has, whether it says so or not, committed to a hidden-variable account of quantum mechanics — and hidden-variable accounts live or die by Bell. This paper stops avoiding the question and answers it, accepting the uncomfortable consequences rather than hiding them, because a clear falsifiable commitment is worth more than a comfortable evasion.

2. Why the Rope Model Is a Hidden-Variable Theory

The programme's ontology holds that each rope carries a real, physical imbalance orientation — a definite direction, around the rope cross-section, in which one strand dominates — whether or not any measurement is made. This is the ontology's central commitment and its principal appeal: it is what lets the programme claim a physical picture UNDERNEATH the quantum formalism rather than merely reproducing it.

The consequence is forced. If the imbalance is real and definite before measurement, then a measurement outcome is determined by that pre-existing orientation together with the measurement setting. That is the definition of a HIDDEN-VARIABLE theory: the hidden variable λ is the imbalance orientation, and the quantum state is a statistical description of an ensemble of definite λ . The rope model does not get to be agnostic here — realism about the imbalance IS a hidden-variable commitment.

This immediately subjects the rope model to every theorem that constrains hidden-variable theories — most decisively, Bell's. The rest of this paper is the confrontation.

3. Bell's Theorem: The Constraint That Cannot Be Evaded

Bell's theorem states that no theory satisfying all three of the following can reproduce the correlations quantum mechanics predicts (and experiment confirms):

1. LOCALITY: a measurement outcome on one side does not depend on the setting freely chosen at a spacelike-separated other side.
2. REALISM (definite values): outcomes are determined by pre-existing hidden variables λ carried by the systems.
3. MEASUREMENT INDEPENDENCE: the hidden variables λ are statistically independent of the freely chosen measurement settings.

For any such theory the correlations obey the Bell (CHSH) inequality $S \leq 2$. Quantum mechanics predicts, and loophole-free experiments confirm, violations up to the Tsirelson bound $S = 2\sqrt{2} \approx 2.83$. The decisive experiments — closing the locality and detection loopholes simultaneously (2015, multiple groups) — leave no room: nature violates the inequality.

The theorem is substrate-independent

Bell’s theorem is a statement about correlations, proven from the three assumptions alone. It makes no reference to what the systems are made of. A theory built from ropes gets no exemption: if it is local, realist, and measurement-independent, it obeys $S \leq 2$ and is falsified by experiment. There is no fourth assumption to relax and no substrate loophole. The rope model must therefore give up exactly one of the three.

3.1 A necessary clarification: non-locality is not a signal

Before proceeding it is worth removing a common confusion, because it colours how readers hear the word “non-local.” A non-local CONSTRAINT is not the same as a propagating INFLUENCE. When we conclude below that the rope model must be non-local, we do NOT mean that a measurement here sends a faster-than-light signal to there. We mean that the two outcomes are bound by a global constraint that cannot be reduced to any local common cause. Quantum mechanics itself is non-local in exactly this sense — it violates Bell — yet permits no signalling. The rope model must reproduce that same combination: a genuine non-local constraint that nonetheless carries no signal (§7.2). Holding these two ideas apart from the outset prevents the mistaken reading that non-local realism means faster-than-light communication.

4. The Forced Choice: Which Assumption Does the Rope Model Give Up?

Since the rope model must abandon one of Bell’s three assumptions, we examine each door and show that two are closed to it, leaving exactly one.

Assumption dropped	What it would mean for the rope model	Verdict
Realism (2)	the imbalance is NOT definite before measurement	CLOSED — guts the ontology; the ropes would add nothing to standard QM
Measurement independence (3)	settings correlated with λ (superdeterminism)	REJECTED — a conspiracy loophole that corrodes the programme’s own falsifiability
Locality (1)	the imbalance is real and definite, settings are free, but correlations are established non-locally through the medium	THE ONLY OPEN DOOR — and the one the rope substrate naturally fits

Why realism cannot be the door

Abandoning realism means the imbalance has no definite value until measured. But the entire purpose of the rope ontology is to supply a real, physical, definite substructure beneath the quantum formalism. A rope model without a definite imbalance is not a physical picture underneath quantum mechanics; it is quantum mechanics with decorative ropes. The programme cannot walk through this door and remain itself.

Why superdeterminism should be refused

Abandoning measurement independence — letting the hidden variables correlate with the experimenters' free choices — can reconcile any local model with any data, which is exactly why it is scientifically corrosive: it can “explain” anything and forbids nothing. A programme whose distinguishing virtue has been strict falsifiability should refuse this escape on principle, not merely on taste. We reject it explicitly.

5. The Commitment: Non-Local Realism

With realism retained and measurement independence retained, the rope model must abandon LOCALITY. We state this as a definite commitment rather than a reluctant concession:

The rope model is a non-local hidden-variable theory

The two-strand imbalance is real and definite (realism); experimenters choose settings freely (measurement independence); and the correlations between distant measurements are established through a genuinely NON-LOCAL connection in the rope medium. In this the rope model belongs to the same family as Bohmian mechanics — an explicitly non-local, realist completion of quantum mechanics. This is a distinctive, falsifiable commitment, and we make it openly.

This commitment is precise and consequential: it makes definite predictions (§7) and exposes definite vulnerabilities (§8), which is what a scientific position should do.

6. The Non-Local Connection: Shared and Linked Rope Topology

A non-local commitment is only credible if the ontology supplies a non-local CONNECTION for it to cash out. The rope substrate does, and this is a genuine point in its favour. When an entangled pair is created, the two systems are not merely correlated by a common past — in the rope picture they are two ends of ONE shared topological structure: a common rope, or a pair of ropes joined by a definite linking, established at the source.

The correlation between distant measurements is then a correlation between two ends of a single extended object, not a signal that propagates between them. Stated for a mathematical reader: the entangled pair is a SINGLE CONNECTED CONFIGURATION subject to a PERSISTENT GLOBAL CONSTRAINT — a topological continuity between the two regions — rather than two independent systems sharing a label. This is the ontological feature that makes non-locality natural

rather than bolted-on: the two ends were never truly separate systems; they are one connected structure that happens to be measured in two places.

We are explicit about the limit of what this section provides. It identifies the ONTOLOGY the non-local connection requires — a single connected rope configuration under a persistent global constraint — but it does NOT provide the DYNAMICAL LAW that updates the joint state when a measurement occurs. There is, at present, no equation here. A skeptical reader is right to ask what governs the shared connection, and the honest answer is that the dynamical law is left to a successor paper (§ Future Work). Identifying the required ontology without yet possessing its dynamics is a real but bounded state of progress, and we prefer to state it plainly than to dress suggestive language as mechanics.

Crucial refinement (this is where it gets hard). A shared structure alone is not enough, and getting this wrong falsifies the model instantly (§8). If the shared rope carried a FIXED pre-set orientation fixed at the source — a classical ribbon given a definite twist at creation — that would be a LOCAL hidden variable, and Bell experiments would already have refuted it. The non-locality must therefore be genuine: the measurement at one end must non-locally UPDATE the JOINT state along the shared rope — constrain it, not causally act on it — while leaving each end’s marginal statistics unchanged. The connection must be more than a shared label — it must be a shared, dynamically-responsive state.

7. The Two Calculations That Decide the Framework

Choosing non-local realism is necessary but not sufficient. To be viable rather than merely consistent, the rope model must pass two quantitative tests. We state them as the concrete work that would confirm or kill the framework, and we do NOT claim either is yet done.

7.1 *Reproduce the exact quantum correlation*

The model must yield the specific correlation of quantum mechanics between measurements at relative angle (a–b):

$$E(a,b) = -\cos(a-b) \quad , \quad \text{with CHSH value } S = 2\sqrt{2} \text{ (the Tsirelson bound)} \quad (7.1)$$

Not merely SOME violation of $S \leq 2$, and not MORE than quantum mechanics (super-Tsirelson correlations are not observed and would falsify the model just as surely as too little). The rope dynamics must reproduce the cosine curve and saturate the Tsirelson bound exactly. The appearance of a COSINE is at least not coming from nowhere: the programme already derives spinors and a Hopf-fibre phase geometry (the two-strand state space is the Hopf bundle, established in the electromagnetic and topological papers), and cosine correlations are the natural language of projections in exactly such spinor/fibre geometries. Whether those structures GENERATE the quantum correlation quantitatively, rather than merely resembling its ingredients, remains to be shown — and is the single most important open problem of this paper. We stress that this is not a technical detail but the paper’s central scientific RISK: many non-local hidden-variable models produce correlations that are neither local nor exactly quantum, and landing precisely on the cosine (and no further) is what the rope dynamics must be shown to do.

7.2 Respect no-signalling

The non-local connection must not permit faster-than-light communication. Quantum entanglement violates Bell yet cannot signal: the marginal statistics at each wing are independent of the distant setting. The rope connection must have the same property — measuring one end must not change what the other end SEES, only how the two outcomes are CORRELATED. A rope mechanism that let a measurement here alter the marginal there would predict signalling, which is not observed, and would be falsified immediately. Proving that the shared-rope connection respects no-signalling is the second necessary calculation, also unfinished.

Why these two together are demanding. They pull in opposite directions: the connection must be strong enough to produce the full Tsirelson-bound correlation, yet constrained enough to leak no signal. Quantum mechanics threads this needle exactly; any completion must reproduce that balance. Bohmian mechanics achieves it by construction. The rope model must be shown to achieve it — not assumed to.

8. The Central Vulnerability: The Local-Disguise Trap

We now state plainly the way the rope model is most likely to be wrong — not to undermine it, but because naming the sharpest failure mode is how a theory earns scrutiny rather than dismissal.

The trap

It is very easy to believe one has a NON-LOCAL rope explanation of entanglement when one actually has a LOCAL one in disguise. If, on close inspection, the shared-rope correlation reduces to a value FIXED at the source and merely carried, unchanged, to both ends — a common cause, a pre-set twist — then it is a local hidden variable. Such a model obeys $S \leq 2$ and is ALREADY FALSIFIED by loophole-free Bell tests. No amount of rope language rescues it.

Avoiding this trap is therefore not optional and not a matter of presentation. The burden is to exhibit a shared-rope mechanism in which the measurement genuinely and non-locally updates the joint state — demonstrably not reducible to a pre-set common cause. Until that is shown, the model has not established that it is on the living (non-local) side of the line rather than the dead (local) side.

We regard this as the honest centre of the paper. The rope model's quantum-foundational viability rests entirely on its non-locality being genuine. That is a demanding, verifiable condition, and we do not claim to have met it — we claim to have identified it precisely, which is the prerequisite for meeting it.

8A. Why Bell Does Not Rule Out a Continuous Medium

Bell's theorem is often discussed in terms of PARTICLES: two separated particles, each carrying hidden variables, measured independently. The theorem does not depend on that picture — it constrains correlations however they arise — but the particle framing shapes intuitions about how SURPRISING non-locality is. The rope ontology is not a theory of particles. It is a theory of a continuous, extended, interpenetrating medium, and this changes how non-locality should FEEL, though not whether it is required.

In a medium picture, two distant measurements are not interactions between two disconnected objects that must somehow coordinate across a gap. They are interactions with two regions of ONE continuous substrate. A constraint linking the two regions is then no more mysterious than the two ends of a taut rope being unable to move independently — the connection is carried by the shared extended object, not signalled across empty space between two points. This does NOT evade Bell: the correlations still violate the inequality, and the medium connection is still, technically, non-local (it is not reducible to a local common cause). What the medium picture offers is not an escape but a reframing: non-locality is what one should EXPECT of measurements on a single connected substrate, rather than a paradox demanding explanation. The surprise in the particle framing — how do two separate things coordinate? — dissolves when the two things were never separate to begin with.

9. Relation to Existing Interpretations

Locating the rope model among established positions clarifies what is and is not novel about it, and prevents it from claiming credit for structures it inherits.

Interpretation	Bell assumption dropped	Relation to the rope model
Copenhagen / standard QM	realism (no definite pre-values)	Opposite choice; the rope model is realist where Copenhagen is not
Bohmian mechanics	locality (non-local pilot wave)	SAME family: non-local realism. The rope model is a substrate candidate for a Bohm-like guiding structure
Many-worlds	none (denies single outcomes)	Different: no collapse, no hidden variables; the rope model keeps definite outcomes
Superdeterminism	measurement independence	Rejected here (§4) as corrosive to falsifiability
Rope model (this work)	locality	Non-local realism with a specific proposed substrate (shared/linked ropes)

The honest reading: the rope model does not invent a new interpretive category. It lands squarely in the non-local-realist family alongside Bohmian mechanics, and its distinctive contribution is a candidate PHYSICAL SUBSTRATE for the non-local connection — not a new resolution of the measurement problem. Its advantage over an abstract pilot wave would be a concrete mechanism; its burden (§7, §8) is to show that mechanism actually works.

10. Measurement and Decoherence in the Rope Picture

The earlier decoherence results now find their interpretive place. In a non-local realist model, measurement is not a separate axiom (as in Copenhagen) but a physical interaction in which the measured rope structure becomes entangled with the many degrees of freedom of an apparatus, and the phase coherence of the isolated system is dispersed into the environment — decoherence, as developed in the earlier phase-coherence paper. The definite outcome corresponds to the definite

underlying imbalance; the apparent “collapse” is the practical irreversibility of that dispersal, not a fundamental process.

The mesh framework can say something more specific than a generic decoherence account, and this is where the interpretation becomes uniquely “rope.” In the thermodynamics paper, the entropy of a macroscopic field is the configuration degeneracy $S = k \ln|C|$. A measurement, in those terms, is an **IRREVERSIBLE INCREASE IN CONFIGURATION ENTROPY**: the measured rope structure becomes correlated with an apparatus whose vast configuration space makes the joint state overwhelmingly unlikely to re-cohere. “Collapse” is then not a fundamental process but the practical irreversibility of that entropy increase — the same coarse-graining the thermodynamics paper treats, now doing interpretive work. This ties measurement to the programme’s own statistical mechanics rather than importing decoherence as an external module.

We flag the limit honestly: this explains why interference is lost and why outcomes appear definite and irreversible, but not, on its own, why **THIS** outcome occurs rather than that one. In the non-local realist reading the answer is that the outcome was fixed by the actual value of the hidden imbalance — which returns the full weight of the framework to the requirement that this hidden value, and its non-local correlations, reproduce the quantum statistics (§7). The entropy-increase account is a consistent and distinctively rope component, not a substitute for that hard calculation.

11. Falsifiability: How This Framework Could Be Killed

The framework is exposed to refutation at several sharp points, which we state plainly:

4. If the shared-rope mechanism is shown to reduce to a pre-set common cause (a local hidden variable), the model is already falsified by loophole-free Bell tests (§8). This is the primary risk.
5. If the rope dynamics cannot reproduce $E(a,b) = -\cos(a-b)$ and the Tsirelson bound — producing too little violation ($S \leq 2$) or too much ($S > 2\sqrt{2}$) — the model is falsified (§7.1).
6. If the shared-rope connection permits signalling (a measurement changing the distant marginal), the model predicts faster-than-light communication, which is not observed, and is falsified (§7.2).
7. If future experiments were to reveal correlations **BEYOND** the Tsirelson bound (super-quantum), that would falsify quantum mechanics itself and require reassessment of any completion, including this one.

We regard risk (1) as the sharpest and most immediate: it is a matter of internal derivation, not of awaiting new data. The model must demonstrate genuine non-locality on paper before it can claim to have survived Bell at all.

12. Status of Each Claim

Claim	Status
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The rope model is a hidden-variable theory	Derived (forced by realism about the imbalance)
It cannot be a LOCAL hidden-variable theory	Derived (Bell + loophole-free experiment)
It must be non-local realist (Bohm family)	Derived (by elimination of the other two doors)
The shared/linked-rope non-local connection	Mechanism sketch (plausible, not derived)
Genuine (non-disguised) non-locality of that connection	Open Problem — the central vulnerability
Reproduction of $E(a,b) = -\cos(a-b)$, Tsirelson bound	Open Problem (unfinished calculation)
No-signalling of the connection	Open Problem (unfinished calculation)
Decoherence as the account of measurement appearance	Consistent (inherited; not a solution to outcome selection)

The distribution is honest for a foundations paper: the NEGATIVE results (local models falsified; non-locality forced) are solid, because they follow from a theorem and experiment. The POSITIVE results (that the rope mechanism actually delivers the quantum correlations non-locally and signal-free) are open. The paper's strength is the sharpness of the forced conclusion; its honesty is the openness of the constructive work.

13. Connections to the Rest of the Programme

- Dirac equation & phase coherence: supply the emergent formalism whose INTERPRETATION this paper fixes as non-local realist.
- Decoherence: provides the account of why outcomes appear definite and interference is lost (§10).
- Topology (Hopf structure, linking): the shared/linked-rope connection (§6) is topological, tying the non-local mechanism to the same linking mathematics that defines charge and the EM sector.
- Thermodynamics: the dispersal of coherence into environmental degrees of freedom is the same coarse-graining/entropy increase treated in the thermodynamics paper.

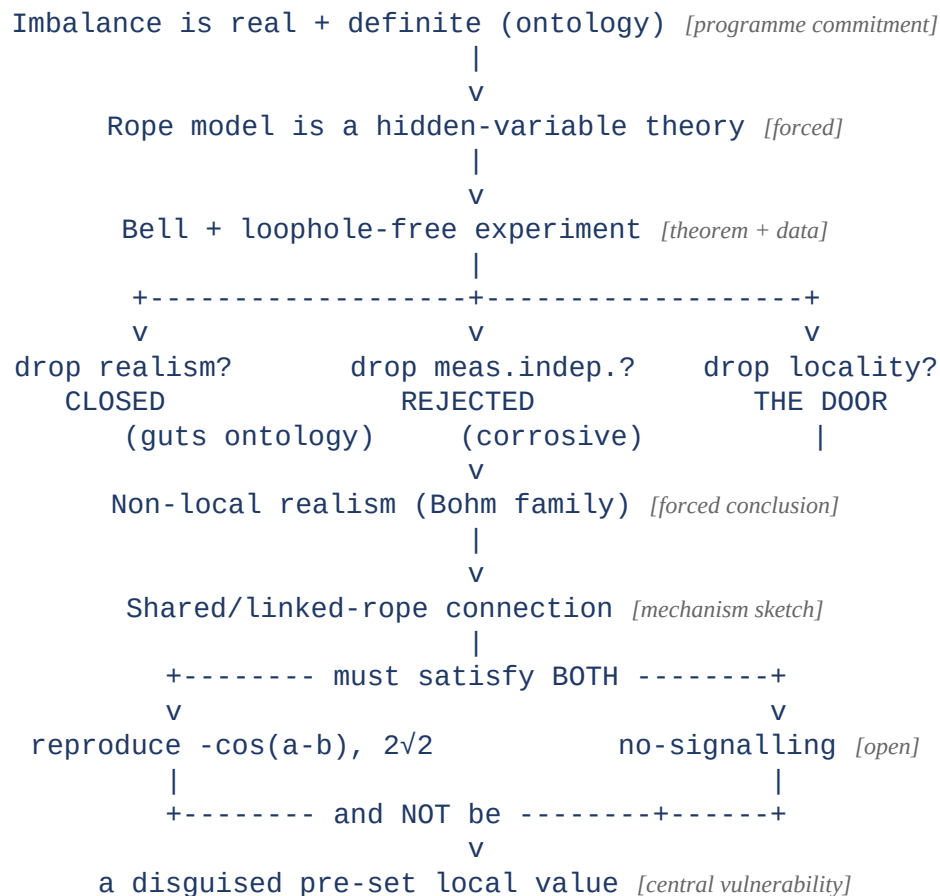
14. Future Work: Dynamics of Non-Local Rope Correlations

This paper identifies the required ontology — non-local realism realised as a single connected rope configuration under a persistent global constraint — but deliberately stops before the dynamical law that would make it mechanics. The natural successor, “Dynamics of Non-Local Rope Correlations,” has a single objective: write the equation that updates the shared-rope joint state under measurement, and show that it (i) reproduces $E(a,b) = -\cos(a-b)$ at the Tsirelson bound, (ii) respects no-signalling, and (iii) is demonstrably non-local rather than a disguised common cause. The programme's existing spinor and Hopf-fibre structures are the natural raw material for that derivation.

We regard the outcome of that successor as genuinely two-sided, and valuable either way. If the rope dynamics DO generate the cosine correlation and no-signalling, the quantum-foundational sector is substantially strengthened — the ontology would have earned its non-locality. If they CANNOT, that failure would locate, precisely, a place where the rope ontology needs revision. A framework that turns an interpretive dispute into a definite calculation with a definite pass/fail is doing what a scientific programme should.

Appendix A. Dependency Graph

The logical structure of the argument, showing where the conclusion is forced and where the burden of proof remains.



The upper half is forced by theorem and experiment; the lower half is the open constructive burden. The single node on which everything rests is the last: the connection must be genuinely non-local, not a local common cause wearing rope language.

Conclusion

Confronting Bell's theorem forces the rope model into a definite position: it is a hidden-variable theory; it cannot be a local one without being already falsified; and since it can abandon neither realism nor measurement independence, it must be an explicitly non-local hidden-variable theory in the family of Bohmian mechanics. The rope substrate offers a plausible non-local connection — a single connected topological configuration established at pair creation — which is a genuine point in its favour. But the framework's viability rests on constructive results it does not yet possess: a derivation of the exact quantum correlation and the Tsirelson bound from rope dynamics, a proof that the connection respects no-signalling, and above all a demonstration that its non-locality is genuine rather than a local common cause in disguise. We have not delivered these; we have identified them precisely and stated the model's central vulnerability without flinching. The contribution is a clear, distinctive, falsifiable commitment at the interpretive core — the place the programme had until now avoided, and the place it is most exposed.

Programme credit: the core Rope Hypothesis is due to Bill Gaede; this work develops and formalises it. Computational collaboration and drafting by Claude (Anthropic). Correspondence to palmer100@gmail.com.